



Fact, Say, or Fiction: Factivity Alternation, Expletive Negation in Korean/Japanese vs. Non-canonical *whether* in English, and Imaginative Worlds

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Abstract

We examine how different complementation types help determine the semantic class of the attitude predicates that select them. We show cross-linguistic variation of factivity alternation, where canonically factive predicates such as ‘know’ and ‘recognize’ develop non-factive alternants—a phenomenon distinct from the inherent non-factivity of doxastic predicates such as ‘think’ and ‘believe.’ These non-factive alternants instead take SAY-related (reportative) complementation, rather than the FACT-selecting complements of their factive uses, and arise only when nonveridical logical or discourse factors—such as temporal distance, misbelief, or counter-doxastic contexts—override the predicate’s default FACT profile. Fictive predicates such as ‘imagine’ also employ this non-factive, SAY-type complementation.

We propose broad cross-linguistic constraints governing attitude expression. We show how surface-ambiguous complements are resolved in different languages by focus or complement morphology. We argue that local perspective-shift accounts do not adequately account for the

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non-factive alternants of cognitive-discovery predicates and emotive factives. These patterns suggest a structural distinction between factive complements headed by a covert FACT and non-factive complements headed by a covert SAY. We also analyze ‘non-canonical’ *whether*-complements with neg-raisable predicates (‘think’, ‘believe’), and with ‘hope’, ‘fear’, and ‘imagine’, comparing them to expletive negation in Korean and Japanese—both functioning as logical expletivization yielding positive bias. Finally, we consider how ‘imagine’ (whether- or that-complements) contributes to the creation of aesthetically unusual worlds.

Keywords: attitude predicates, factivity alternation, perspective shift, global alternation view, expletive negation, polar question, noncanonical *whether*, positively biased, hedge, fictive.

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1. Introduction¹⁾

This paper seeks to explore how attitude predicates are related to

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their embedded complement contents. We will see how different complementation types determine their higher attitude predicate types. We try to show how cognitive/semantic structure, syntactic structure, and prosodic structure are correlated and further show how languages vary in factivity alternation (FACT-related) for ‘know’ and other epistemic predicates, vary in reportative (SAY-related) complementation for doxastic predicates such as ‘think’/‘believe,’ and vary in FICTION-related complementation for fictitious predicates such as ‘imagine.’ We thus seek some universal constraints regarding attitude expressions.

In 2, factivity alternation will be demonstrated. It will be shown how the alternation occurs in English by means of nonveridical operators such as interrogative, negation, conditional, modal, and *before* ---, indicating how pragmatic perspective shift view is limited and local. Then, it will be shown how factivity alternation in Korean occurs by means of complementizers (as in all Altaic), whereas Chinese rarely shows factivity alternation except in the verb ‘remember.’

In 3, we further see semantic grounds for epistemic predicates and factivity, analysing the ProFactNoun *kes* (K) and *koto* (J), along with related issues.

In 4, we discuss how logically trivial embedded complements can be if the complement content contains either expletive negation as in Korean and Japanese or non-canonical *whether* as in the set of predicates such as {think, believe, hope, fear} (*whether*). Both of them are actually psychologically real and (positively) bias-oriented with hedge (디딤말) (and even politeness).

In 5, we discuss fictitious or imaginative worlds and related expressions, conjecturing how they can be well represented by means of fiction-related predicate ‘imagine’ in various languages, dealing with non-metaphysical time/world changes.

6 concludes the paper, looking forward to universal semantic/pragmatic principles of factivity.

2. Factivity Alternation Patterns and Theoretical Issues

Some attitude predicates carry a factivity presupposition. The use of English *know* presupposes the factivity of its complement clause, as in (1). Knowledge-related cognitive attitude predicates similar to *know* may be classified as epistemic predicates. These attitude predicates all carry a factivity presupposition of their complements, which is not cancelled or suspended, so it is projected, even when negated (capitals: focus).

(1) English *KNOW* presupposes the factivity of its complement clause:

- a. Mary KNOWS that Ken left. $\sim \rangle$ Ken left
- b. Mary DOESN'T KNOW that Ken left $\sim \rangle$ Ken left

However, unlike the regular (anti-veridical, see Zwartz 19), entailment-cancelling ($\neq \Rightarrow$) but commitment-preserving, use of negation as in (1b), if 'not' is used as a weaker nonveridical propositional operator, as in (2), the presupposition of the embedded complement content is cancelled (or not projected). As a consequent, *know* here becomes non-factive, and factivity alternation occurs for the cognitive predicate 'know' in English. Then, the attitude holder 'Mary' in (2) comes to have the doxastic attitude roughly similar to 'not believe that Ken left.'

(2) Mary doesn't know that Ken LEFT.²⁾ $\sim/\sim \rangle$ Ken left.

Such nonveridical propositional operators in English include interrogative, modal, conditional, and *before* in English. With those applied, epistemic attitude predicates similar to *know*, such as

2) This non-factively negated *know* is realized in Korean by means of a non-factive complementizer.

recognize, discover, find out, realize, be aware, see, and understand, may become non-factive, as factivity alternation predicates. For (3), Horn (2014) notes that the complementizer that is necessary for its non-factive reading. This point is contrasted with the deletability of the presupposed complement from (1a): ‘I know(, I know)’ or ‘I know it/that’ with the anaphor referring to the Theme complement. In (4a), the conditional factive predicate with focus but with no focus in its complement content (CC), the CC projects, whereas in (4b), with narrow focus (i.e., focus of a constituent smaller than a clause) in the CC of the conditional cognitive predicate, the CC does not project. In the conditional of (4b), the CC of the epistemic predicate has narrow focus on **PLAGIARIZED**, the interpretation of which contributes to a set of alternative propositions (Rooth 1992). In consequence, the cognitive factive predicate *discover* becomes non-factive (Beaver 2010). In (5), the modal *may* (or perhaps at the S head) functions as the alternation operator, making the sentence non-factive. In (6), ***before* is another nonveridical operator and does not allow factivity alternation; it licenses only non-factive reading of *know* (or *realize*).** E.g. Ken *realized* that he was late *before* the meeting started. Here, the complement’s truth is evaluated in a prior world-time. Thus, in English, a fixed number of nonveridical operators with focus positions (in complements from (2) to (6) above) license non-factive alternants of cognitive factive verbs.

- (3) Do you know that he’s RELIABLE? (Hintikka 1972) (focus mine)
- (4) a. *If* the TA **DISCOVERS** that your work is plagiarized, I will be forced to notify the dean. (see Beaver 2010, Mahler et al 2020)
 b. *If* the TA discovers that your work is **PLAGIARIZED**, I will be forced to notify the dean.
- (5) He *may* have understood that our garbage isn’t being DECOMPOSED.
- (6) Koreans knew (*the fact) that CHINESE CHARACTERS were the best *before* Hangul was invented (see Hazlett 2010).

From (7) below, we consider a discourse/pragmatic view of *know* and other factive predicates such as *recognize* and *be glad/regret*; yet both English and cross-linguistic data show that under certain discourse-pragmatic conditions, these predicates admit non-factive alternants. One approach is the ‘perspective shift’ view – called ‘protagonist projection’ (Holton 1997; Stokke 2013; Abrusán 2020), ‘viewpoint shift’ (Hinterwimmer 2017), or ‘perspective shift’ (Harris & Potts 2009). On this account, the epistemic base of ‘know’ shifts from the narrator’s global common ground to the protagonist’s local belief state. The alternative view, which I propose here, treats factivity alternation as a matter of presuppositions being either projected globally or suspended globally (and confined to subordinate contexts). The perspective shift view is regarded as a cousin to *free indirect discourse* (FID) but is distinct. (7) below, can be approached by Hinterwimmer’s (2017) ‘viewpoint shift’: Instead of shifting the temporal indexical ‘six hundred years ago,’ the epistemic modal base for *know* shifts. Abrusán (2020), finding Gazdar’s (1975) early treatment, offers an analysis using Eckart’s (2014) (and Holton’s) story update where Protagonist Perspective updates the context with ‘x (the perspective holder) sees,’ for his/her epistemic/doxastic state but not for the Common Ground of narrator and addressee, sentence-internally at the level of finite clauses ([CP ...] embedded by *know*), based on eventuality. The available epistemic modal base for *know* shifts locally. At the given time point, a universally quantified set of humans *knew the fact that the Earth was flat*, which is not true, globally viewed.

- (7) a. In the Middle Ages, everyone “knew” (*the fact)/ that the Earth was flat.³⁾

3) Pat Clancy, via e-mail (Nov. 13, 2024), “The “scare quotes” (ironic and the like) on the verb clearly indicate that it is being used in a meaning that differs from its actual (or

- b. Six hundred years ago, everybody knew the world was flat, and if you sailed too far, you would drop off the edge. (Google)
- (8) a. [[She knew that he would never let her down]]^{M,g<C,d>}, but, like all the others, he did. (Holton 1997)
- b. I just KNEW that would happen! [said after a small accident like spilling coffee] #But that didn't happen. Michael Barrie (p.c.)
- c. [[She KNEW that he would never let her down]], #but, like all the others, he did.
- d. [[She KNEW that he would never let *me* down]], ##but, like all the others, he did. (Manfred Krifka (p.c.). Because of *me*, I assume that it contributes to forming common ground.)

Here in English, contextually determined discourse/pragmatic factors license non-factive alternant interpretation of factive verbs. For (7a) above, as Pat Clancy (p.c., Nov. 13, 2024) observes, the scare quotes clearly indicate a non-factive, ironic reading. Without them, the sentence is judged infelicitous. Craige Roberts (p.c., Nov. 24, 2024) concurs: without scare quotes, the presupposition is false; but with scare quotes or as Clancy indicates, in a narrative context immersing the reader in medieval beliefs ("everyone knew that there were four humors and that the Earth was flat"), the sentence becomes acceptable. The scare quotes mark that *know* here is used in a meaning distinct from its canonical factive sense.

Instead of shifting the temporal indexical 'six hundred years ago,' the epistemic modal base for *know* shifts. Abrusan (2021, 2020), finding

'factive') meaning." Craige Roberts (Nov. 24) agrees on this. Without the scare quotes, they regard the sentence infelicitous. "I could imagine a text in which the author immerses the reader in a story that takes place in the Middle Ages and then mentions some things everyone believed in those days, e.g., "Everyone knew that there were four humors and that the Earth was flat." (Pat) So, the preceding text brings the reader into **the point of view** of the people of the time, and the reader understands without explanation." Pragmatic/discourse view of non-factive interpretation of 'know.'

Gazdar's (1975) early treatment, offers an analysis using Eckart's (2014) (and Holton's) story update where Protagonist Perspective updates the context with 'x (the perspective holder) sees,' for his/her epistemic/doxastic state but not for the Common Ground of narrator and addressee, sentence-internally at the level of finite clauses, based on eventuality. The available epistemic modal base for *know* shifts locally. The scare quotes (ironical) reading for most implies that it is not factive but non-factive. The author from the external context must show up to have control of the entire situation and reflect scientific truth. The (scare) quotes implies that it is just a quotation from others. This coincides with complement marking to get a non-factive alternant reading of *knew* [+say] *that the Earth was flat*, where the [+say] complementizer, corresponding to *-ko/-n kes-uro* 'reportedly/from thinking' in Korean, as will follow. In Korean and other Altaic languages, the ProFactNoun is employed for the factively presupposed complement and the [+say] complementizer for its non-factive alternant.

Furthermore, even for English, the factive alternants of (8b) and (8c)&(8d) have their stressed focus on the verb *knew*, not on any constituents of their complements. This should add up to the factivity alternation operators for the factive predicate *know* in particular. Here the embedded futuristic modal of certainty *would* should be crucial. By alternation, if the factive verb *know* happens to be non-factive, as in (8a), the contradictory utterance can follow grammatically, but in contrast, as in (8c) (or (8d), if the epistemic verb *know* is interpreted as in its factive alternant reading (with focus on the verb), the following contradictory utterance with *but* creates anomaly. The alternation-marking language Korean clearly shows its alternation by complement-endings ProFact Noun for factive alternant *kes* vs. the non-factive alternant [+say]/Reportative C *-ko/kes-uro*.

In contradictory contexts, the basic epistemic factive verb *know*, discovery factive verb *realize*, and the emotive factive verb *regret/glad*

all can be followed by a presupposition denial clause, as we saw already: *Ken {**knows, realizes, regrets**} that it is thundering outside, #but it is not thundering outside.* But even here Korean (and most Altaic languages), can mark both factive vs. non-factive alternant readings of the verbs involved and if the complements end in non-factive endings, the following *but* clause can follow without anomaly.

Even factive verbs of discovery and emotive factive verbs, after (9) for *know* as the basis, (10) and (11), can be non-factive by the Protagonist's misconception: Their Korean counterparts are marked as non-factive grammatically, with factive-marking irredeemable. First, examine the contrast between *know* and discovery factive verbs in the the false belief context, which is distinct from the contradiction context, which we observed above.⁴⁾ We will first examine the factive verb *know* in (9a) in English to see how it is distinct from discovery and emotive factive verbs in false belief contexts, * and # co-existing with no formal distinction in English, unlike in Korean, where the non-factive alternant has a distinct complement ending with no violation. English is compared with Korean and Japanese. In Japanese, 'know' has an accidental gap of no non-factive alternant of 'know' (so (9b'') is lacking) and Chinese has no factivity alternants of discovery factives (except independent, separate 'think', 'believe' type verbs), as noted below. (10) is about discovery factive verbs and (11) is about emotive factive verbs, followed by Korean (b), Japanese (c), and Chinese (d) examples. In (9a'), we can see the violation of Ross's (1967 [1965]) factive island constraint for the factive verb *know*, which has no non-factive alternant with any false belief context, unlike predicates of discovery and emotive predicates, as in (10) and (11), respectively.

4) Paul Kerr, our language consultant, made the distinction between a and b clearly, though some failed to. Remember the contradiction context with *know*: *Ken **knows** that it is thundering outside, #but it is not thundering outside.* The same happens with *realizes* and *regrets*.

- (9) a. */#Falsely believing that he had inflicted a fatal wound, Ken *knew* that he was a murderer.⁵⁾
 a.' *Who did Ken know that he was a murderer of?
 b. Ken-un caki-ka chimyengcekin sangche-rul iph-i-ess-ta-ko
 K -TOP self -NOM fatal wound-ACC inflict-CAUS-PST-DEC-C
 calmot mit-ko, caki-ka sarinca-i-n kes-*ul//kes-uro
 al-ass-ta
 falsely believe and self-NOM murderer-COP-DEN-ProFactN-ACC/thing as
 k n e w
 *knew/[non-factively knew or justifiably believed]
 'Ken falsely believed that he had inflicted a fatal wound, and he
 {*knew that he was a murderer/'non-factively knew' that he was
 a murderer.' With *kes-ul*_{ProFactN-ACC}, contradictory *, with *kes-uro*
 'regard as,' ✓(no violation). [K]
- b.' __, nwukwu-rul Ken-un[caki-ka _ sarhay-ha-n kes-*ul//kes-uro
 who-ACC K-TOP self-NOM murder-do-ADN Pro-FactN-ACC /thing-as
 know-Q
 'Who does Ken know that he killed __?' [* island violation with
 Pro-FactN-ACC COMP but ✓(no island violation) with *kes-uro* 'regard
 as' COMP ending].
- c. *Ken-wa zibun-ga aite-ni timeisho-o owaseta-to omoikonde
 K-TOP self-NOM other-DAT fatal wound-ACC inflict-[+say]C falsely
 thinking, zibun-ga satsuzinsha-ni natta-koto-o shi-ta
 self-NOM murderer-TO became-Pro-FactN-ACC -know-PST
 'Ken, falsely thinking that he inflicted a fatal wound to someone, knew
 that he became a murderer.'
- c.' * ---omoikonde, dare-o zibun-ga koroshita-koto-o shi-ta?
 falsely.thinking who-ACC self-NOM killed-ProFactN-ACC know-PST
 '*Falsely thinking, who did he know that he killed.' (*(contradictory) factive
 island)

Chinese, as an isolating language, syllable-morpheme units tend to form lexical units and factive-only verbs *yishidào* 'realize,' *rèn-chū* 'recognize' and *fā-xiàn* 'discover,' including *zhidào* 'know,' have no non-factive alternants and cannot be in this alternation position. Instead, independent non-factive only verbs of *yǐwéi* 'think'/'believe' and *chǎn-shēng* 'give rise to (doubt, etc.)' are used (for the last verb, Linmin Zhang, p.c.). Consider *zhidào* 'know':

5) Marking (9a) with #, the intuition of those who "rescue" it pragmatically is captured and why some tend to accept it as a factive alternant is understood. But the logical contradiction is risked.

- d. wù yǐwéi Lǐ shā-le rén, *Bǎo zhīdào Lǐ shì shārénfàn. [C]
falsely believe Li kill-PERF person Bao know Li be murderer
'Falsely believing that Li killed someone, Bao knew that Li was a murderer.'
- d.' wù yǐwéi -, *shéi, Bǎo zhīdào [Lǐ shā-le t_i]?
falsely believing -, who Bao know Li kill-PERF
'---, *Who did Bao know that Li killed?' *(factive island violation)
- (10) a. Falsely believing that he had inflicted a fatal wound, Ken **realized** that he was a murderer. (from Gazdar 1979)⁶
a.' Ken had never killed anyone. #However, he realized that he was a murderer.
a." Falsely-, who did Ken realize that he was a murderer of?
✓ (no island violation)
- b. Ken-un caki-ka chimyengcekin sangche-rul iph-i-ess-ta-ko
K -TOP self-NOM fatal wound-ACC inflict-CAUS-PST-DEC-C
calm not mit-ko, caki-ka sarinca-i-n kes-uro insikhay-ss-ta
falsely believe and self-NOM murderer-COP-DEN-thing as
'Falsely believing that he had inflicted a fatal wound, Ken recognized(*realized) that he was a murderer.' DEN=DENOMINAL [K] ✓
- b.' -, nwukwu-rul Ken-un caki-ka cwuk-i-n kes -uro insikha-ni?
who-ACC K-TOP self-NOM die-CAUSE-ADN thing-AS recognize-Q
'-, who does Ken recognize that he killed?' [K] ✓ (no island violation)
- c. Ken-wa zibun-ga aite-ni timeisho-o owaseta-to omoikonde
K-TOP self-NOM other-DAT fatal wound-ACC inflict-[+say]C falsely thinking,
zibun-ga satsuzinsha-ni natta-to ninsiki-si-ta(*ki-ga tuita)
self-NOM murderer-DAT became-[+say]C recognized(*realized)
'Falsely believing that he had inflicted a fatal wound, Ken recognized(*realized) that he was a murderer.' [J]
- c.' --, dare-o Ken-wa zibun-ga _ korosita-to ninsiki-si-ta-ka?
who-ACC K-TOP self-NOM kill-[+say]C recognized?
'Who did Ken recognize that he had killed?' ✓ (no island violation)
- ! Nota bene: Just like 'realize' *kkaytat-* in K, *ki-ga tuku* 'become aware, realize' in J functions as a factive verb only, whereas both K *insikhi-* 'recognize' and J *ninsiki-* 'recognize' can be non-factive alternants.
- d. wù-yǐwéi zìjǐ shā-le rén, #Bao yishidào [zìjǐ shì shārénfàn]. [C]
falsely believing that self kill-ASP someone B recognized self be murderer

6) My post hoc glance at Zhu's (2025) analysis of *recognize* and *regret* is impressive and may be generally in line with my global view.

'Falsely believing that he killed someone, Bao recognized he is a murderer.'

- d. ' wù-yǐwéi zìjǐ shā-le rén, *shei, Bao yìshìdào [zìjǐ shā le t_i].
false believe self kill someone who Bao realize self kill PERF
'Falsely believing that he killed someone, who did Bao realize that
he killed?' [C]

- (11) a. Jane misheard that the dinner was free and she was **glad** not
to have to pay. (after Karttunen 2016)

- b. Ken-un cenyekpap-i kongcca-i-ra-ko calmot tut-ko ton-ul an
K -TOP dinner -NOM free-DEC-C be falsely hear-Conj money-ACC not
nay-to toyn-ta-ko **coha-hay**-ss-ta
pay-though all right-DEC-[+say]C glad-DO-PST-DEC [K]
'Ken misheard that the dinner was free and he was glad not to have
to pay.'

! Nota bene: After a false belief clause context, in particular, the
[+say]C *-ko* (or *-kes-uro* 'regard as') clause is attached and some
reason (not necessarily a fact) for the expressed emotion, as an
adjunction, seems to be given. This may also apply to the following
Japanese example with the same [+say]C *-to* clause in the emotive
expression. This may also apply to the English and Chinese
correspondents, though with no formal distinction for the
non-factive rendition. In K and J, however, a separate factively
marked NP/DP C clause is possible, in which case, with the false
belief clause, a contradictory semantic violation occurs, e.g.,
*--- ton-ul an nay-to toy-nun kes -ul_{ProFactN-ACC} (or *-kes-ey*
tayhay 'regarding the fact that-').

- c. Noa-wa dina-ga muryo-da-to kiki-matigae-te
N-TOP dinner-NOM free-DEC-C hear.err-ger do-ger
harawa-naku-te ii-to yorokonda
pay-NEG-ger OK[+say]C rejoiced [_{ger=geundive}]
'Noa, mishearing that the dinner was free, rejoiced at her not
having to pay.' (based on Yosinari Harada p.c.)

- d. Li yiwei wancan shi mianfei de, ta hen gaoxing ziji buyong fu qian.
L (mis)believe dinner was free he was glad he didn't have to pay
'Li misbelieved that the dinner was free and he was glad not to have
to pay.' (Chinese: confirmed by Linmin Zhang p.c.)

This kind of mis-perception clause or gerundive construction easily suspends (or cancels) presupposition for those particular factive kinds of verbs and indeed their factivity is pragmatically suspended (or cancelled for *know* in English). The Protagonist in various contexts like (7b), (8a), (9a), (10a), and (11a) may be sympathetic with the local cognizer/experiencer but the non-factive status must be clearly revealed by means of the author and common ground, as in my global view. For them, factives are uniformly factive and presupposition failures are patched up by shifting to a joint set of narrator + protagonist. In our global view, the local view may be represented by presupposition triggers possibly in presuppositional DRSs (π), to be resolved (suspended) in the global DRS or a subordinate one. In belief/attitude contexts: a DRS K_att (attitude DRS) is embedded under a predicate like 'believe' [Basic DRS: $K = \langle U, Cond \rangle$, where U is a set of discourse referents and $Cond$ a set of conditions]. Thus, those non-factive alternants of discovery and emotive factive verbs as well as *know* can be shown in DRS.

K0: [e, Ken, x]	Ken falsely believes K1

K1: [x]	Ken inflicted a fatal wound on x
K2: [e']	Ken realizes [he was a murderer]
=====	

Fig.1 DRS in false belief contexts (discovery factive alternant *realize*) .

Following EH Lee's (2020) distinction between event time and discourse-anchored evaluation time, we may assume that K_1 contains its own temporal anchor tA_1 (e.g., '600 years ago') without affecting presuppositional anchoring. This allows K_1 to represent both the subject's local belief p ('the Earth is flat') and the historically distant temporal context in which that belief is situated, while presuppositions continue to be evaluated only in K_0 ('flat(Earth)' non-factively).

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K0: [t0, e, x, p]
    ¬p (author CG: not flat(Earth))
    ∀x(person(x) → K1)
-----
K1: [e']
    know(x, p) & Time(e') = tA1
    p: [ ] flat(Earth)
=====

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Fig. 1' DRS Interpretation: K_0 evaluates presuppositions at t_0 ($\neg p$), while K_1 introduces its own temporal anchor tA_1 , modeling the subject's distant-time presuppositional belief p .

Local participants at discourse anchored time tA_1 ('600 years ago') treated p as true. This creates a mismatch: 'know' is factive only in the local context, but non-factive from the global perspective. The finger quotation marks around "knew" can reflect this context-dependant pragmatic non-factive usage.

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K0: [e, ken, p]
    mishearing(e1, ken, p)
    p: [ ] dinner_free
K1: [e2]
    rejoice(e2, ken, q)
    q: [ ] not_have_to_pay(ken)
-----

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Fig. 1'' DRS: The non-factive alternant of an emotive factive verb

The truth of the embedded proposition is only locally accessible in K_1 , not globally entailed. In order to represent the non-factive alternants in K, J and other Altaics in a fair manner, we need a renovated DRS as follows:

K_0	Author / Common Ground <i>recognize_f</i> (p presupposed, entailed) <i>recognize read</i> (take as p)
K_1	Subject's presupposition p: 'the portrait is <i>real</i> ' (false belief context $\neg p$)
K_2	Imaginative world

Fig. 0 Factivity Alternation and Imagination across Discourse Layers (DRS Format).

Non-factive readings are author-readable in K_0 without being globally entailed; a distinct world ($w_0 \neq w_1$) arises not only with imaginative predicates.

How about in other languages such as Korean and Chinese in genral? In Korean (and other Altaic languages in general (Lee, C. 2019)), center-embedded complement clauses of epistemic verbs take either a nominalized *ket* '것' (ProFactN), followed by Accusative form *-ul* '-을' as a factive alternant ending, or a [+say] Reportative complement form *-ko* '-고' (or *ket-uro* '것-으로') as a non-factive alternant endig.⁷⁾

7) Therefore, 'Some Koreans still *know* that Chinese characters are the best, although they are not used any longer in public' can hardly be ambiguous. Because of 'some', accoring to them, it may be a fact. However, the narrator with common ground reflects public opinion and common sense, making it non-factive. This narrator's or global view applies to the following as well: 'Some Koreans *knew* that Chinese characters were the best even after Hangul was invented.'

Chinese, on the other hand, needs different verbs of ‘think, believe’ to get a non-factive sentence.

(10') Hankwukin-tul-un [Hangul-i changcey-toy-ki cen-ey-nun]

[hanca-ka Korean-PL-TOP K alphabet-NOM create-become-Nmnlzr

before-at-TOP C characters-SUBJ choiko-i-n kes-uro] al-ass-ta

best be-C-thing-as know-PST-DEC

‘Koreans, before Hangul was invented, regarded Chinese characters as the best.’

In Chinese, non-factive readings can be represented only by a different doxastic verb **rènwéi** / **yǐwéi** ‘think/believe’ and the cognitive factive verbs of ‘know,’ ‘recognize,’ etc. cannot be employed. The verb ‘remember’ **jì-dé** alone is an exception.

(11') zài Hángwén fāmíng yǐqián, Hánguórén { **rènwéi** / **yǐwéi** / * **zhīdào** }

at Hangul invention before Koreans **think believe know**

Háan-zi` shì zài-hǎo de.

‘Before the

invention of Hangul, Koreans thought Chinese-characters were the best.

The interplay between prosodic patterns and factivity

Although Chinese doesn’t allow factivity alternation in most epistemic verbs including ‘know’, the verb ‘remember’ **jì-dé** shows alternation. **Focus position** can be a crucial factor leading to factivity alternation in the use of the surface-structurally ambiguous **attitude predicate jì-dé** (‘remember’) in a written text.

(12) Lǐ-Sì jì-dé [jīn-tiān shì fā-xīn-rì

Lǐ-Sì remember [today is payday].

- A. Factive reading: Lǐ Sī REMEMBERS that today is payday.
If JIDE is focused, ‘today is payday’ becomes presupposed.

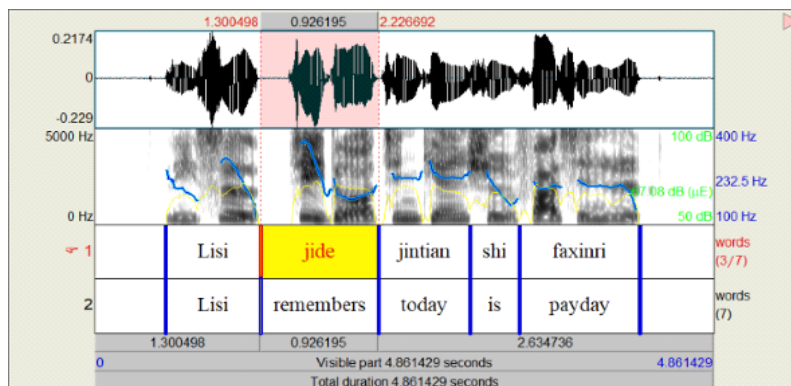


Fig. 2 The attitude predicate JI-DE ‘remember’ is focal and high, bearing focus.
(A Mandarin female Chinese speaker at 32 recorded this.)

- B. Non-factive reading: According to Lǐ Sī’s memory, it’s TODAY that is payday. (A complement constituent is focus)
C. |= TODAY is payday.

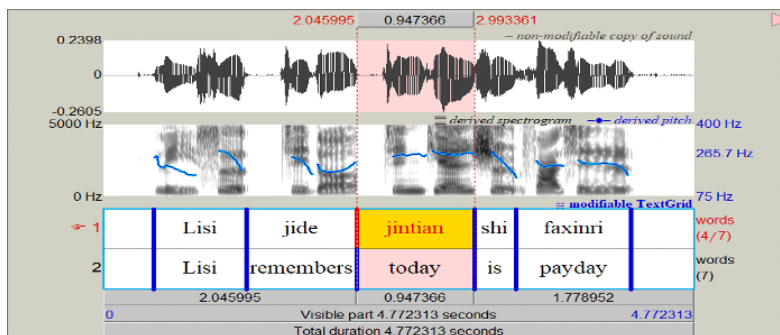


Fig. 3 The complement part JINTIAN ‘today’ is focal and high, bearing focus.

As shown in (12a) below, when the attitude predicate *ji-dé* is focused itself, the sentence has a factive reading, i.e., with a factivity entailment. Here what is at issue is whether Lǐ Sī remembers or not, and what is remembered (i.e., the complement part) is backgrounded and factively presupposed. In other words, (11’a) expresses the memory of a fact. As shown in (7), there can even be an overt

apposition, *shì-shí* ('the fact that').

D. Lǐ-Sì jì-dé_F jīn-tiān shì fā-xīn-ri (**de shì-qíng / shì-shí**).

Lǐ-Sì remember today is payday **of** thing / **fact**

= (11a): Lǐ-Sì remembers the fact that today is payday. **Factive**

However, as shown in (11'b) below, here although syntactically, the whole sentence remains the same, and prosodically, the focus falls on an element within the complement clause, i.e., *jīn-tiān* ('today'). This focus position change (dubbed as '**compensatory prosodic focus marking**') brings a semantic consequence: there is no factivity entailment. As shown in (8), for this non-factive reading in (11b), there can be an optional *shuō* ('say') after the attitude predicate *jì-dé* ('remember'). (See also Section 3 for related discussion.)

(13) Lǐ-Sì jì-dé-(**shuō**) jīn-tiān shì fā-xīn-rì.

Lǐ-Sì remember-**say** today is payday

= (11'b): According to Lǐ-Sì's memory, it's today that is payday.

Non-factive

Then in Korean, as illustrated in (10) and (11), usually the use of case markings provides clear clues on whether there is factivity entailment. In (10), the use of the accusative marker, *-ul*, along with the abstract noun *cwul* indicates that the complement clause is interpreted as a fact, and there is factivity entailment. In (11'), the use of the postposition *-lo/-uro* ('as') removes factivity entailment. Although the same predicate *al-* ('know') is used in (11') as in (10'), (*cwul-lo*) *al-* has rather the meaning of 'believe justifiably (with evidence)' (see more discussion in Section 3).

(14) Ken-un [Mia-ka ttena-n cwul-**ul**] **al**-a

Ken-top Mia-nom leave-c ProFactN-ACC know-DEC

Factivity entailment: 'Ken knows that Mia left.' |= Mia left

(15) Ken-un [Mia-ka ttena-n cwul-**lo**] **al**-a

Ken-TOP Mia-NOM leave-c N-as know-DEC

No factivity entailment: 'Ken believes with evidence that Mia left.'

Mia left

In colloquial (or written) Korean, case markers or prepositions like *-ul* and *-lo* can be optionally deleted. If deleted, then again focus position can determine whether there is factivity entailment, as illustrated in (14). When it is the predicate *al-* that is focused, factivity is entailed (see (14a)); when it is part of the complement, here *Mia*, that is focused, there is no factivity entailment (see (14b)).

- (15) a. Ken-un [Mia-ka ttena-n cwul-Ø] al-a
 K-TOP M-NOM leave-PstC ProFN know-PrsDEC
Factivity reading: ‘Ken knows that Mia left.’ |= Mia left
- b. Ken-un [Mia-ka ttena cwul-Ø] al-a
 K-TOP M-NOM leave-PstC noun know-PrsDEC
Non-factivity reading: ‘Ken believes with evidence that Mia left.’
 |=/= Mia left cf. Jeong (2020), which lacks structural correlations.

5

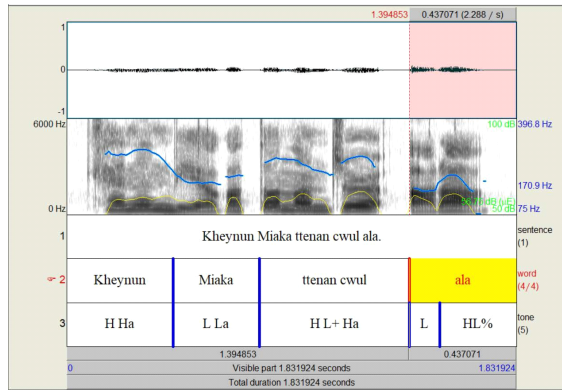


Fig. 4 The S-final attitude verb AL-A ('know') is focal and high, bearing focus, when the ACC marker is deleted from the ProFactN cwul in DP (12a)

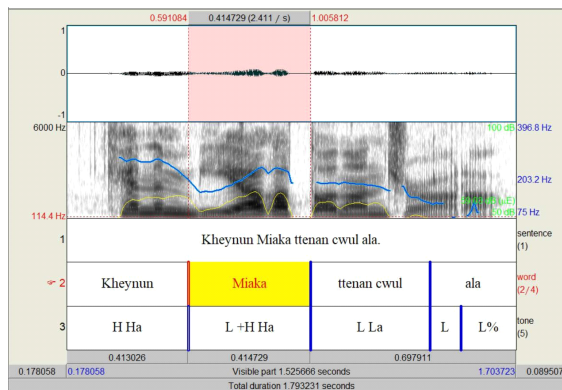


Fig. 5 The complement part MIA-KA ('Mia') is focused and high, bearing focus

To sum up, cross-linguistic data suggest a robust pattern: focus position indicates what an underlying QUD is and what alternatives are taken into consideration (see e.g., Rooth 1985, Roberts 1996/2012), which in turn determines the availability of factivity entailment. When it is the attitude predicate that is focused, the complement clause becomes backgrounded and its factivity is presupposed. When it is part of the complement clause that is focused, the alternative set of the complement clause would be taken into consideration, and there is no factivity entailment.

Our position is that factively presupposed complements in Korean and Japanese (with other Altaic languages) appear in truncated nominalized structures (with ProFactN). This suggests that the factive alternant is structurally simpler, not more complex. By contrast, in languages like English and Chinese, where complement structures are surface ambiguous, presupposed complements require compensatory focusing on the factive verb, while non-presupposed complements require focusing on the complement itself. Eye-tracking and other processing studies indicate that non-factive alternants or variant non-factive verbs in such languages are more costly to process (e.g., Chinese *jide* ‘remember’). These findings, when publicized, challenge de Cuba’s (2007) assumption that the factive verb complement is more complex. Instead, factive nominal (DP) complements with truncated forms or even surface ambiguous factive alternant should be taken as simpler than full finite complements. Cross-linguistic data and further experimental work will be crucial in deciding where the real complexity lies---whether complements in the factive or non-factive alternants/verbs.

3. Meanings and Factivity

3.1 Factivity Alternation and Neg-Raising

In Korean and other Altaic languages, the non-factive alternants from epistemic predicates can undergo neg-raising, as shown in (16), where the negation in the higher embedding clause in (16a) is in the lower embedded clause in (16b) and still (16a) is interpreted as (16b), both basically identical in meaning. It is because the non-factive alternants are **doxastic** just like *mit-* 믿다 ‘believe.’ If *kes-uro* 것으로 is changed to *kes-ul* [ProFactN-ACC] 것을 in (16a, b), as factive alternants, no neg-raising relations occur because they are epistemic or factive. In Korean and Altaic in general, the aspectual form *-ko iss-* ‘and be,’ attached to the epistemic verb *al-* ‘know,’ is stative (although it becomes progressive if attached to an action verb).

- (16) a. Ken-un [Mia-ka ttena-un kes-uro] al-ko iss-ci anh-ta
 K-TOP M-NOM leave-PstC thing as know-STATIVE-C not [neg-raised]
 건이는 [미아가 떠난 것으로] 알고 있지 않다.
 ‘Ken does not ‘nonfactively know’/justifiably believe that Mia left.’
 b. Ken-un [Mia-ka ttena-ci anh-un kes-uro] al-ko iss-ta
 K-TOP M-NOM leave-C not-PstC thing as know-STATIVE-DEC
 건이는 [미아가 떠나지 않은 것으로] 알고 있다.
 ‘Ken ‘nonfactively knows’/justifiably believes that Mia didn’t leave.’

How about in English? In English, the same neg-raising relations are attested in non-factive alternants from epistemic predicates but only if they belong to the eventive class veridically defined, rather than presuppositional, such as *prove/demonstrate/show/establish*, unlike *know, recognize, discover, etc.*

Observe (17b) for neg-raising below. Here, the eventive verb *prove* undergoes neg-raising but if it is replaced by a non-eventive,

presuppositional *discover*, neg-raising fails.⁸⁾ (17) below shows that the lower negation in an eventive veridical cognitive predicate *prove* clause (17a) is in the neg-raising relation with its higher embedding negation clause (17b).

- (17) a. If the T.A. proves (#discover) that your work is **NOT** ORIGINAL, I will be forced to notify the dean.
 b. If the T.A. does **not** prove (#discover) that your work is ORIGINAL, I will be forced to notify the dean.

Thus, both in Korean and English, the non-factive alternants of factive predicates (more broadly in Korean) undergo neg-raising, just as in antirogatives like ‘think,’ ‘believe’ (see 5).

3.2. Content Nominal Heads Are Consistent with Clausal Complements in Factivity

Content DPs may be divided into two: the factive ones such as [*the fact* [*that P*]] and non-factive ones such as [*the rumor* [*that P*]]. A factive DP denotes a unique entity such that the entity is a fact in w and the content of x in w_o , as shown below.

- (18) [[the fact that Mia left]]
 $= \iota x[\text{fact}_w(x) \wedge f_{\text{cont}}(w)(x) = \lambda w_o. \text{Mia left in } w_o]$ (see Elliott 2016)

- (19) [[The rumor that Mia left]]
 $= \iota x[\text{rumour}_w(x) \wedge f_{\text{cont}}(w)(x) = \lambda w_1. \text{Mia left in } w_1]$

8) Ozyildiz, D. indicates that Lee, C. was the first to find neg-raisability in non-factive alternants of the epistemic factive class. He is refining the condition. Whether the embedding predicate is eventive or stative matters. Cf. Zeretic & Ozildiz (20).

(20) Ken knows [_{DP} the rumor that Mia left]. (Ken is familiar with 'the rumor,' involving acquaintance with an entity of type ⟨e⟩, without commitment to its truth, resolving Vendler's (1972) paradox of *know*, not entailing the complement clause, which modifies its direct head). Our head and content consistency works). This contrast in content nominals foreshadows the distinction between factive and non-factive CP complements, particularly where [+say] heads are overt with their adnominal finite clauses with [+say]C in Altaic.

Nota bene! *-ko*: In Altaic, the center-embedded CP has its finite T P / I P / M o o d P (declarative/interrogative/imperative/proposative (/promissive) clause) followed by covert SAY, bleached by REPORT C, which signifies non-factivity, composing with attitude verbs. The C *-ko* may have a light 'say' structure under C or simply a [+say] feature assigned to C, bleaching veridicality. Even embedded under 'know' or other factive predicates, the proposition is still **non-factive** as an alternant, as we already examined in the alternation section.

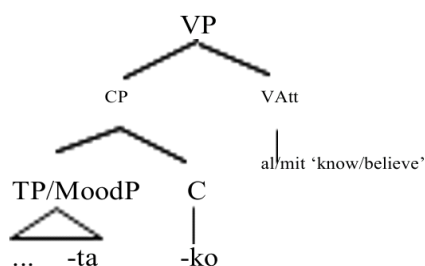


Fig. 6 [+say]C(omplementizer)

- (21) a. Ralph believes [say] Ortcutt is a traitor. (see Kratzer 2013)
- b. [[say]] $\lambda p \lambda s \forall w (w \in f_{\text{content}} \rightarrow \exists S' (S' \leq w \& p(s')))$.
 $p = \text{traitor}(\text{Ortcutt})(s')$
 (i) Propositional argument: p .
 (ii) Event or situation argument: s , the modal anchor.
- c. Content modality: f_{content} maps entities that determine intentional content to the set of possible worlds that are compatible with that content (Hacquard 2006; Kratzer 2006; Moulton 2009)
 Kratzer introduces a covert [say] as the head of the complement clause of *believe* in English. In German, a subjunctive mood (with *se*) is employed in its complement.

3.3 Non-factive Nominals: all involving [+say] complement

- (22) a. [_{PRO} [Mia-ka ttena-ss-ta]-ko HA-nun] somwun/cwuchang-i
 Mia-NOM leave-PST-DEC-[+SAY]C say-ADN.PRES rumor claim-NOM
 'the rumor/claim that Mia left'
 phe-ci-ess-ta
 spread-get-PST-DEC
 'The rumor that Mia left spread.'
- a' *[Mia-ka ttena-n somwun]-i phe-ci-ess-ta
 Mia-NOM leave-ADN.PST rumor
 (Intended) 'The rumor that ⟨factive ADN⟩ Mia left.'
- (23) a. — ta to IU uwasa/hanashi (Japanese) [parallel with 22a in K]
 -PST QUOT say rumor/news (slightly modified from Iida 2018)
 a' *— ta uwasa -PST rumor [parallel with (22a') in K]

All speech act nominals (commands, questions, proposals, promises, propositions) are inherently non-factive, because they encode illocutionary force rather than veridical content, involving [+say] complement in Altaic languages. Note that all the different finite

mood markers appear, as sentential endings in the complement clauses. This is different from English and other Western languages.

(24) speech act nominals – non-factive (in K and J)

- a. na-ka-ra-(ko ha-)(*to iu* in J)nun myengryeng)(-ul hay-ss-ta)
 out-go-IMP-([+say]C SAY-ADN order-ACC do-PST-DEC
 ‘(() gave) an order [[+say] imp] to go out’ (*to iu* in J)
- b. nwu-ka w-ass-nunya-(ko ha)nun cilmwun(질문)(*to iu* in J)
 who-NOM come-PST-Q-SAY-ADN question
 ‘a question saying (asking) who came.’
- c. ttek-ul mek-ca-(ko ha)nun ceyan(제안)(*to iu* in J)
 cake-ACC eat-PROP-[+say]C say-ADN proposal
 ‘a proposal saying let’s eat cake’
- d. moca-rul sa-cwu-ma-(ko ha)nun yaksok(약속)(no parallel in J)
 hat-ACC buy-give-PROM-[+say]C promise
 ‘a promiss saying I promiss to buy you a hat.’

- (25) hanul-un phwuru-ta(ko ha)nun myengcey(명제)
 sky-TOP blue-DEC([+say]C say-ADN proposition
 ‘a proposition saying that the sky is blue.’

A ‘proposition’ is something debatable in public and is inherently non-factive.⁹⁾ All such speech act nominal forms are non-factive, since they encode illicutionary force rather the veridical content.

3.4 Topic position also licenses factive content (not only Theme role)

Nominals in topic position behave differently depending on factivity. A factive nominal such as ‘the fact that P’ can be evaluated as correct, but not as incorrect, due to presupposed truth.

- (26) a. #[The fact that Mary left] is not correct/right.
 b. # [Mary-ka ttena-n **kes-** -un] mac-ci anh-ta (same in J)

9) Elliott’s (2016) opposite view seems misleading. ‘Proposition’ in ‘propopositional content’ must be distinct.

M-nom leave^{-PSTC} ProFN-TOP right^{-C} NEG-DEC
 '[The fact that Mary left] is not right.'

(26') a. [The fact that Mary left] is correct/right.

b. [Mia-ka ttena-n kes-un] mac-ta

M-NOM leave^{-PST} ADN ProFN TOP correct^{-DEC}
 'The fact that Mia left is correct.'

(27) a. That Mary left is not correct/right.

b. [Meari-ga sat-ta-to iu koto-wa] tadashi-ku-nai

Mary-NOM leave^{-PST-C} say ProFN-TOP right^{-NEG}
 'That Mary left is not right.' (to *iu*: same with *-ko ha* in Korean)

Unlike the nominal fact head, clausal topics with reportative C (*-ko ha* / *to iu*) can felicitously take negative evaluative predicates like incorrect.

Likewise, we can see that factive presupposition is licenced not only under factive predicates as their Theme complements.

3.5 Public/Communicated Knowledge

As Wittgenstein (1969) observed, a child saying (28) means 'I have learned it,' a case of public, communicated knowledge, not an internal discovery.

(28) I know that the Earth turns round.

Korean and Japanese distinguish between internal knowledge and external/public/communicated knowledge: the reportative *-ko ha/to iu* layer signals communicated factivity, while bare *-nun kes* ProFactN, signals a stronger, personalized form of internalized factivity. In this particular context, the factive by personal cognition sounds a bit odd (as in (29b), unlike the external factive in (29a)):

- (29) a. na-nun [cikwu-ka tol-n-ta-(ko ha-)nun kes]-ul a-n-ta
 I-TOP earth-NOM turn-PRS-DEC [+say]C do-PRS ProFactN-ACC know-PRS-DEC
 나는 지구가 돈다-(고 하-)는 것을 안다
 'I know that the Earth turns round.' [external/public kind of factive]
- b. ?na-nun [cikwu-ka tol-nun kes]-ul a-n-ta
 I-TOP earth-NOM turn-PRS ADNC ProFactN-ACC know-PRS-DEC
 ?나는 지구가 도-는 것을 안다.
 'I know that the Earth turns round.'

In Korean and Japanese, we can see that (29a) is external or public kind of factive because of -고 하- [+say] or Reportative Complementizer is inside of the outer -(nu)n kes ProFactN factive element. Without the [+say]C involved, it is internal factive (and becomes a bit odd, as in (19b), if you don't observe its turning around, as a spaceman or expert. Thus Korean and Japanese distinguish between epistemic authority grounded in personal cognition and that grounded in public report. In English, the factive *explain* seems to have the external or public kind of factive, whether or not it has its DP head or not (see Elliott's (2016)).

3.5 The ProFactNouns *kes*(것) and *koto*
 Internal pro-fact content nouns such as Korean *kes* and Japanese *koto* contrast with their equivalent lexical nouns meaning 'fact' (*sasi*, *jujitsu*).

- (30) a. [Korean *kes*] (ProFactN) = [Japanese *koto*] (ProFactN)
 b. [Korean *sasi*] 'fact' = [Japanese *jujitsu*] 'fact' = [[English *fact*]]
 $a=b= \lambda w. \lambda x f_c. [\text{cont}_w(x f_c)(w) = 1 \wedge \text{fact}_w(x f_c)]$ (from Moulton's (2015))
 subscript content x_c type entity to subscript factive content type $x f_c$

to make distinction (or simply $[np \text{ } kes] = \lambda x.\text{fact}(x)$)

We treat ProFactN as encoding presuppositional and veridical content (Kim 2009). It is the core, and let's further consider its cognitive aspect. *R* as 'a suitable acquaintance relation' has been specified as $\langle \text{witness-based-perceptual/internalized} \rangle$ here and the condition that *s* is (unique and familiar) in *C* (Context) is added. Here we model 'remember' as requiring a unique, familiar situation in context that grounds the acquaintance relation.

- (31) [the verb 'remember' *ji-dé* [kes/kotoR]]⁹*C* = $\lambda w.\lambda p.\lambda s.\lambda x.R(x)(p)(s)(w)$ (cf. Kim 2009) (*s* is in the domain of situations; *R* is a suitable witness-based-perceptual/internalized acquaintance relation defined iff *s* is unique and familiar in *C* (*g*: model; *C*: context); *x* is an entity (attitude holder); *p* is a complement proposition; *w* is a maximal evaluation situation)

If the *-ko HA/- to IU* 'SAY' layer above the finite TP/IP/MoodP content clause precedes, the content is in the Common Ground via communication or public proposition. The conditions here apply to all the equivalent content nouns *sasil* (Korean) = *jujitsu* (Japanese) = *shi-shi* (Chinese) = *fact* (English), as given in (80).)

We treat Korean *-(n)un* and Japanese pitch rise as complementizers on a par with English *that*. We can thus carry over the semantics of the clause-initial [*that*] in English to our clause-final prenominal modifying formative *-(n)un* in Korean and its corresponding prenominal modifying pitch, not falling (see Lee 1989, Venditti et al. 1996 for experimental support).

- (32) [[Korean (*n*)*un*]]
 = [[pitch rise (Japanese)]]
 = $\lambda p \lambda x_c \lambda w [\text{cont}(x_c(w)) = p]$
 (applying Kratzer 2006 and Moulton 2015)

Thus, we can have the PreNomF *-kes* (K) or *koto* (J), attached to the prenominal modifying formative *-(u)n* (K) or its equivalent pitch contour ‘not low’ (J) as a C, to form $\langle -(u)n \text{ } kes \rangle$ (K) or $\langle \text{not low pause } koto \rangle$ (J), which will be preceded by a truncated part of a factive complement clause, becoming equivalent to *the fact (overt/covert) that P* in English.

All epistemic predicates such as *recognize*, *be aware*, and *remember* entail $[[\text{know}_f]]$ and their factively presupposed complements – instantiations of covert or overt [fact] – may be composed with these epistemic predicate **in the mental lexicon**. Emotive factives such as *regret* and *be glad* felicitly require ‘know(x, p).’ In Korean and Japanese, the element [fact] may be realized by pro-factive nominalizers *kes* and *koto*, respectively.

Lexically negative, anti-veridical verbs that entail $[\text{know}_f]$ such as Korean $[[\text{moru-}]]$ ‘not know,’ $[[\text{ic-}]]$ ‘forget’), Chinese $[[\text{bù } \text{`-jìdè}]]$ ‘not remember,’ $[[\text{wàng-jì}]]$ ‘forget’), are likewise derived from $[[\text{remember}_f]]$ (Lee 2019). By contrast, syntactic (clausal) negation – as opposed to lexical negation – may yield non-factive readings in Altaic languages and in English.

The verb *forget* illustrates the split clearly. As a lexical factive $[[\text{forget}_f]]$, it presupposes its complement. But it lacks a non-factive alternant, unlike its positive counterpart $[[\text{remember}_f]]$, which alternates with $[[\text{remember}_{nf}]]$. As an implicative $[[\text{forget}_{imp}]]$, however, it loses its presuppositional force: the complement is no longer presupposed but rather forms part of the asserted or entailed event — as in “He forgot

to lock the door,” in its infinitival, which implies that he did not lock it. The implicative variant, whether in *remember* or *forget*, is restricted to an infinitival complements in E and involves volitional modality, patterned after action verbs. This contrast is robust cross-linguistically.

In Korean, *kiekha-* ‘remember’ is factive ~ non-factive alternating with ProFactN *kes* vs. [+say]C *ko* (or *kes-uro* ‘regard as’) but is weakly implicative. Examine (33a, b) in K, where *-nun* is Present/Constant/Normative to produce an implicative, whereas *-ul* is Future/Modal to weaken an implicative. The Japanese *wasureru* ‘(forget_f)’ (or ‘forget_{impl}’) shows tense differences; (34a, b) in J, where *-u* is Present with no Future/Modal element possible to produce a solid implicature; $[[wasureru_{impl}]](p)(x) \Rightarrow \neg p(x)$. In J, *no* is equivalent to the PreNomF *koto* for factivity and is employed for implicative more often than *koto*. In C, *bù-jide_{impl}* or *wàngji_{impl}* ‘forget’ as implicative typically may need modal *yào* ‘need’; interrogative and imperative sentences with *jīde* ‘remember’ are fully implicative, and its declarative is weakly implicative. The implicative regularity in (35) in C is cross-linguistically attested, although its declarative *jīde* implicative is rather weak.

- (33) a. Mia-nun mwun-ul camku-**nun**/-**ul** kes-ul ic-(y)ess-ta [K]

M-TOPdoor-ACCclose-ADNC-//MOD/FUTthingACCremember/forget-PST-DEC

(Intended) ‘Mia remembered/forgot to close the door.’

- b. Mia-nun mwun-ul camku-nun/-ul kes-ul kiekhay-ss-ta

M-TOPdoor-ACCclose-ADNC-//MOD/FUTthingACCremember/forget-PST-DEC

- (34) a. kusuri-o nom-**u** no-/koto-o wasure-ta => did not drink

medicine-ACC drink-PRES drink-NMNLZR-ACC forget-PST

‘() forgot to drink medicine.’

- b. kusuri-o nom-**u** no-/koto-o oboe-te iru => did drink

medicine-ACC drink-NMNLZR-ACC remember-STATIVE

- () remember to drink medicine.’ (Y. Harada, O. Sawada)
- (35) a. Li **bu**-jide/*wàngjì* [(yao) $\emptyset$ suo hao *damen*] => Li bu suo hao *damen* le.
 L not remember/forget must lock well door Li not lock well door ASP
 ‘Li didn’t remember/forgot to lock the door.’ => Li didn’t lock the door.
- b. Li jide [(yao) $\emptyset$ suo hao *damen*] (?) => Li suo hao *damen* le.
 (Intended) ‘Li remembered to lock the door.’
- b.’ Li *wàngjì* [(yao) $\emptyset$ suo hao *damen*] ma? => Li bu suo hao *damen* ma?
 b.” *Wàngjì* [(yao) $\emptyset$ suo hao *damen*]! => Bie suo hao *damen*!
 ‘Forget to close the door!’ => Don’t close the door!

The analyses of factive alternation and implicative predicates show that meaning composition in attitude contexts depends on the interaction between lexical entailment type, complement structure, and the [fact] operator. Factivity is not just a lexical property but a compositional outcome: epistemic and emotive predicates entail $[[\text{know}_f]]$ and select complements marked by [fact] or its morphophonological realizations such as Korean *kes* or Japanese *koto*, and their non-factive alternants via [+say]C in complements, requiring compositional procedures, while implicative readings arise when the [fact] feature is suppressed by modal or infinitival morphology. Thus, presupposition projects when [fact] is realized, but entailment emerges when it is not.

Across languages, the implicative pattern correlates with volitionality and control: infinitival or nominalized complements referring to intended or unrealized actions yield entailments such as $\neg p$ (‘did not lock the door’). This cross-linguistic regularity confirms that implicativity is event-structural and modal, not epistemic. Consequently, $[[\text{forget}_{impl}]]$ and $[[\text{remember}_{mpl}]]$ encode assertive entailments, while $[[\text{forget}_f]]$ and $[[\text{remember}_f]]$ encode presuppositional knowledge. Factivity and implicativity therefore occupy points along a single continuum: Presuppositional $\rightarrow$ Entailing $\rightarrow$ Volitional.

4. Embedded questions with expletive negation in Korean and noncanonical *whether* in English --- A Parallel

Unlike Responsives (Factives, like *know*, *recognize*, *remember*), Antirogatives (those originally not allowing interrogative complements such as *think*, *believe* and Preferentials such as *hope*, *fear* and *imagine* in addition here) are newly claimed to appear with *whether* interrogative complements in English (White 2021). Similarly, expletive negation with polar interrogative complements in Korean/Japanese is well established, and we try to explain this commonality in terms of the semantics and pragmatics of their complements.

- *Know whether*. We begin by assuming the semantics of *know whether* as (36).

(36) The proposition 'Ed knows whether P' is true iff

i) if P is true then Ed knows P and ii) if P is false then Ed knows not P.

(37) The proposition 'Ed knows that P' is defined when P is true, if defined, then it is true iff the set of possible worlds where P is true is the subset of the set of possible worlds compatible with Ed's doxastic states.

(38) By presupposition: I know whether P or not $\Leftrightarrow$ (P is presupposed $\wedge$ I know that P) $\vee$ ($\neg$ P is presupposed $\wedge$ I know that $\neg$ P) [Note that *whether* is reduced to that].

Logically, accepting bivalence and excluded middle, *know whether* can be defined from *know that*, as in (37). Epistemic verbs are veridical because they fit the definition by entailment (see (37)), but

{think, believe, imagine} *whether* fails to. This set also fails to be factive in terms of presupposition (Kiparsky & Kiparsky 1970, Karttunen 1971b, a.o.). They do not constitute interrogative acts either. Thus, predicates like {think, believe, hope, fear, imagine} *whether* are nonveridical, licensing NPIs.

For English, we focus on predicates such as think, believe, (imagine), hope, fear, including experimental data, arguing that these predicates can take complements with ‘non-canonical’ *whether*. For Korean, we compare those with preferentials (e.g., ‘hope,’ ‘fear’) and doxastics (e.g., ‘think,’ ‘believe’), which we argue can take polar interrogative complements with expletive negation.

Rogatives with interrogative complements in English

The following inquisitive kind of *whether* should be the right example:

(39) Ed asked/wondered **whether** it rained last night.

We will see whether the Korean (and Japanese) expletive negation complements can take a higher interrogative illocutionary act predicate well or not.

White (2021) shows ‘preferential’ predicates with *whether* (hope *whether*, fear *whether*, etc.) and anti-roatives with *whether* (think *whether*, believe *whether*...) are all acceptable by corpus investigation, though some ‘prescriptive’ or ‘academic’ (or ‘canonical’) grammar don’t accept them. Examine (40):

- (40) a. I’m **trying** to think **whether** I’d have been a star today or not. (‘d modal attached)
 b. I was **hoping whether** you are able to guide me ---.

Online Survey: Degree of Acceptability of think, believe, hope, and fear + whether complements. [Survey Date: January 8th, 2024, Respondents: 60 participants, Target Group: Native Monolingual English Speakers, Method: Online questionnaire (via Google Prolific Forms)]

Verb + Wh-complement clause	Freq. of Acceptability							Mean	s.d.	Q1	Q3
	1	2	3	4	5	6	7				
1. I'm trying to think whether	3	12	1	5	14	18	7	4.617	1.85	2.75	6
2. I tried to think whether	0	2	1	3	4	21	29	6.133	1.2	6	7
3. We can choose to believe whether	0	1	2	7	2	19	29	6.05	1.25	6	7
4. I was hoping whether	21	18	3	3	4	8	3	2.783	2.01	1	4.25
5. there is no need to fear whether	3	4	7	12	9	13	12	4.783	1.76	4	7

Table 1. Degree of Acceptability For the Verb-Whether Complements

The acceptability results show that the think whether clause, believe whether clause, fear whether clause were generally rated as highly acceptable(mean = 6.13, 6.05, 4.78, respectively). The findings suggest that those formerly-claimed-to-be-antirogativeverbs are well perceived to take a closed interrogative clause, which is in line with the conclusion drawn from corpus data in White (2021).

These predicates are typically eventively expressed in English and we argue that the complements tend to be positively (likelihood-wise) biased, although White (2021) is not specific about their interpretations.

On the other hand, Korean counterparts (anti-rogatives and preferentials) allow an embedded interrogative complement with *-u*/(modal) and *-kka* (alternative Q), and negation *anh-*, as in (41).

In Korean, complements are surely positively biased, with negation clearly expletive ((Choi Y. and C. Lee 2017). Examine expletive negation examples in Korean and Japanese:

(41) a. Mia-nun [caki thim-i iki-ci anh-ul-kka (ha-ko)]

M-TOP [self team-NOM win-C NEG-MOD-Q/
sayingkak-ha-n-ta/?mit-nun-ta/kitayha-n-ta/siph-ess-ta/sa

n g s a n g h a - n - t a }
 ‘Mia thinks/believes/expects/had a hunch/imagines whether her team will win.’

b. Mia-wa [jibun no chimu-ga katsu node-wa-**nai-ka**-(to)]
 M-TOP [self’s team-NOM win-C-TOP-NEG-Q-[+say]C
 omou/shinji-ru/kitai-suru <no overt modal involved>
 thinks/believes/expects do ‘Mia thinks/believes/expects that her team will win.’

Korean responsiveness allow an interrogative complement with *-ul-kka* (*-kka* as polar Q operator, functioning roughly like ‘whether’) but distinct from *-(nun-)ci*, a real interrogative complementizer just like *whether*, which Japanese lacks) and negation *anh-*(41). In Korean and Japanese, complements are positively biased more obviously than in English. Examine.

Cf. F. Je crains [que vous **ne** prenez froid].
 I fear that you ExpNeg catch-subj.pres cold
 ‘I fear that you may catch a cold.’

(42) a. Mia-nun [nambukhan-i thongiltoy-ci anh-ul-kka (ha-ko)]
 M-TOP North South Korea-NOM united-C NEG-MOD-Q [+say]C
 {sayngkakha-n-ta/?mit-nun-ta/kitayha-n-ta/siph-ess-ta/sangs
 angha-n-ta.} think/believe/expects/had a hunch/imagine
 ‘Mia thinks/believes/expects/had a hunch/imagines that Korea will be united.’

b. Mia-wa [nan-kan-poku-chōsen ga tōitsu sareru kamoshire-nai
 M-TOP South North Korea-NOM unified might/possibly-NEG
 to ommotte iru.
 [+say]C think_{ASP} -
 ‘Mia thinks that South and North Korea might be unified.’

We argue this *anh-* in (41a) is (syntactically) required but semantically expletive. In our view, these indicate a positively-biased thinking with a hedge. With *whether-or-not*, the negative choice is opened, leading to hedge (often even politeness). In (41a), the speaker (attitude holder) desires the addressee's guiding and being a star *irrealis* in subjunctive mood. In English, the semantic sense of disjunctive choice *whether- or-not* still retains in some contexts. In Japanese, as in (42b), a modal kind of expression 'possibly' can appear, although we miss a covert modal auxiliary marker in J, in contrast with K, as in (41b).

The use of complement expletive negation in Korean can achieve the same effect of positively-biased attitude. In (41a), the expletive negation is followed by the conjectural/bouletic modal *-ul* and then the embedded interrogative mood marker *-kka* with the optional [+say] C *-ko* (signaling non-factive, in all Altaic) following, although typically the [+say] C deletes in Korean. The exact parallel occurs in (38b) for Japanese and (41a,b) for Korean and Japanese. This also suggests that White's *whether*, as well as the expletive negation in Korean, makes the complement logically trivial.

We can formalize this bias by means of the best function in bouletic modals (overt in Korean, covert in Japanese, and covert/overt in English): modal terms as follows:

(43) For any $w', w'' \in W$: $w' \prec_{\text{Bou}_x(w_0)} w''$ iff w' is more desirable according to $\text{Bou}_x(w_0)$ than w''

• $\text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0)) = \{w': w' \in \cap \text{Dox}_x(w_0) \wedge \neg \exists v (v \in \cap \text{Dox}_x(w_0) \wedge v \prec_{\text{bou}_x(w_0)} w')\}$

= the set of x 's belief worlds that are best according to the ranking by $\text{Bou}_x(w_0)$

From (42) [positive bias] North and South Korea is desired (hoped/likely) to be united

= $\lambda w_0. \forall w \in \text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0))$ [North and South Korea will

be united in w_0]

On the other hand, factive attitude predicates like *know* and *remember* in their factive alternants in English, Korean and Japanese, presuppose factivity and are incompatible with these ‘think’ type predicates in (41, 42). In the unacceptable *anh-(ass)-ul-kka*, the use of (i) expletive negation *anh-*, (ii) past tense *-ass-*, and modal *-ul*, and (iii) question complementizer *-kka* is incompatible with factive predicates *al-* ‘know’ and *kiekha-* ‘remember’. In contrast, in the acceptable *anh-ass-nunci(-rul)*, the negation is not expletive but contributes to the expression of a negative fact with bipolar-Q-C. Korean factive predicates are compatible with the embedded interrogative **complementizer** *nunci* ‘whether,’ as in (44a) below. J largely lacks this **complementizer**. In K and J, the expletively negated complements end with Q marker, optionally (but not in the complement fronting¹⁰) followed by [+say] Reportative Complementizer, as in (44) below and (42) above, with higher doxastic and preferential predicates.

(44) a. Mia-nun [caki thim-i iki-ci anh-ass-nunci(-rul)]

M-TOP her team-NOM win-C NEG-PST-BiPolC(ACC)

{*al-/kiekha*}n-ta.

know/remember}-PRS-DEC

‘Mia knows/remembers whether her team didn’t win.’

b. Mia-nun [caki thim-i iki-ci #anh-ass-ul-kka

M-TOP her team-NOM win-C NEG-PST-MOD-Q

{ a l - / k i e k h a - } n - t a .

k n o w / r e m e m b e r - P R S - D E C

‘Mia remembers/knows her team didn’t win?’

10) I owe this observation in Japanese (but it also applies to Korean) to Koji Mineshima, who was in my talk.

Our survey recruiting 60 native English speakers supports White's (2021) acceptability claim based on corpus data, but he has no claim on interpretation. We argue the canonical view of attitude predicates that they select different fixed complementizers on logical/semantic grounds must be jettisoned. The parallel between the English noncanonical *whether* and Korean/Japanese expletive negation compel us to consider psychological factors involved: positively biased but (negatively) hedge-built (*whether* also involves *or not*). The complement content matters, determining compatible predicates: a favorable event of her team's winning in the complement selects *hope* and *expect* as embedding attitude predicates, whereas an unfavorable event of her team's losing or catching cold may select *fear* or *worry*. Both kinds of events can be composed with *think* and *believe* in English but *mit-* 'believe' in Korean is slightly odd. Even in English, *believe* is hardly eventive and weaker in positive bias. We can further formalize this bias in terms of bouletic modals ('*d* in E and *-(u)l* in K), which make *anh-* expletive.

This bias could be formalized in terms of bouletic modals overt in Korean, covert in Japanese (but see (42b')), and covert/overt '*d* in English and (44) can simply show the bouletic modal proposition, which is equated to the best function in (43) above.

Under the bouletic modal *-ul*, with *-kka* (Q), NEG *anh-* behaves as the ι -operator in inquisitive semantics, changing the set of sets of possible worlds simply into the set of possible worlds. That is, the negative marker *anh-* under the bouletic modal is expletive. Thus the interrogative illocutinary force of the complement content is lost in Korean, Japanese, and English.

Sudo and Uegaki (2017 The anti-rogativity of non-veridical preferential predicates) argue that even preferential predicates such as *hope* and *fear* cannot take *whether*.

Complement types are important and determining. Goodhue &

Shimoyama (2023, in review) oppose our expletive negation analysis (Choi & Lee 2017), saying that the negation here is not expletive but virtually real. But if another negation is added, double negation law is violated, as in (42), making it a real negative complement and further the complement intonation does not go up as high as its corresponding high negation question does. Furthermore, the anaphor *caki* or *zibun* in the complement, as in (42a,b), is impossible in an interrogative illocutionary act. Crucially, the expletive negation complement cannot be embedded by any interrogative illocutionary act predicate, as shown in (43).

The embedded complement content with expletivized negation matters, with its positive bias determining compatible predicates: a favorable event of her team's winning in the ExplNeg complement selects 'hope' and 'expect' as embedding attitude predicates, whereas an unfavorable event of catching cold is feared by ExplNeg alone, determining its embedding higher predicate fear, instead of hope. Then, how can we express negatively biased content? We need one more negation. Thus, Korea's being not united is worried by double negation, as in (45), and Mia's team losing the game is also worried by **double negation**. Logically, **double negation** is equivalent to positive statement and we use double negation expressions often in a slightly weaker claim (*ani-i-n key ani-ra* 'It's not the case it's not ---').

(45) Mia-nun [thongil-i **an** toy-ci **anh**-ul-kka] yemnye-ha-n-ta

M-TOP [unification-NOM not become-C not-MOD-Q worry-do -PRES-DEC

'Mia worries whether the unification will not (by double neg)
be done.'

b. #Mia-nun [caki thim-i iki-ci anh-ul-kka hako] Ken-eykey mwul-ess-ta.

M-TOP self team-NOM win-C not-MOD-Q [+say]C K-DAT ask-PRES-DEC

‘Mia asked Ken whether her team will win.’

We can also question whether our expletive negation complement can take a higher question-embedding illocutionary act predicate. It’s odd and this is a direct counter-example to Goodhue and Shimoyama’s claim. Observe.

- (46) a. #Mia-nun [caki thim-i iki-ci anh -ul-kka hako] Ken-eykey
M-TOP self team-NOM win-C NEG-MOD-Q [+say]C K-DAT
m w u l - e s s - t a
ask-PST-DEC
‘Mia asked Ken whether her team will win.’
- b. #Mia-wa Edo-ni [nankan-pokuchōsen-ga tōitsu.sareru kamoshire
nai-ka(-to)] M-TOP E-DAT South.Kor.-North.Kor.-NOM unified-POSSIBLY-NEG-Q
[+say] shitsumon-shita.
a s k - d i d
(Intended) ‘Mia asked Edo whether South and North
Korea might get unified.’ (with expletive negation)

In (44), the negation is not expletive but contributes to the acceptable *whether*-like Complementizer in K with a negative fact with alternative interrogative Complementizer *nunci-* ‘whether’. In (45), the unacceptable *anh-(ass)-ul-kka* in its expletive negation form cannot be composed with higher factive predicates *al-* ‘know,’ and *kiekha-* ‘remember’ at the end in SOV

From (42) and (45), we can see that complement negation with Q marker is completely expletivized in Korean and Japanese and can be freely composed with higher cognitive doxastic attitude predicates of antirogatives originally not allowing interrogative complements such as ‘think,’ ‘believe’ and preferentials such as ‘hope’ and ‘fear.’ It can also be composed with a fictitive predicate of ‘imagine’ (think about

different worlds) as well.

But in English, the process of expletivization of *whether (or not)* complements for the embedding preferential, anti-rogative, and fictitive predicates such as *think, hope, imagine* is still on the way socially and historically. Socially, academics accept those rarely. Our survey respondents were all monolingual native speakers. This process is analyzed by way of survey statistics with 60 native speakers employed twice (planning one more survey with those with college education or not). Historically, White's data seem to be rather recent and corpus data such as BNC or American English corpus don't show many old *whether* sentences we seek.

This unified perspective explains typological uniformities across languages and provides a foundation for the next section on imaginative predication, where propositional attitude shifts no longer concern epistemic reality but rather the construction of possible or fictional worlds.

5. Imaginative Worlds

In English, *think*, *believe*, *hope*, and *fear* were thought to be anti-rogative and to take only *that* complement clauses and that they don't take complement clauses with polar interrogative complementizer *whether* until White's (2017) corpus data of *think* and *hope* kinds, as we saw, and Peterson's (2017) dissertation example with *imagine whether* (48) below. The two findings opened a new era of rigorous research in this direction. Naturally I add *imagine whether* and the corresponding Korean and Japanese expletive negation complements to the list.

Compare the following ungrammatical *imagine whether* sentences with the grammatical ones in English and Korean.

- (47) a. *John is *imagining whether* a woman is dancing. (Liefke 2021)
 b. John is *imagining why, how, and with whom* a woman is dancing.
- (48) I am *imagining whether* a new sofa will fit into my living room.
 (Peterson 2017)
- (49) a. say sofa-ka ungcepsil-ey ewulli-l-ci sangsanghay-po-n-ta
 new sofa-NOM living room-into fit-MOD-PollntC imagine-try-PRS-DEC
 'I try to *imagine whether* a new sofa will fit into my living room.'
- b. say sofa-ka ungcepsil-ey cal ewuli-ci **anh-ul-kka**
 new sofa-NOM living room sangsanghay-po-n-ta
 'I desire a new sofa may well fit into my living room.' (positive
 b i a s)
- c. #say sofa-ka nay apat-ey cal ewulli-ess-nun-ci
new sofa-NOM my apartment-into well fit
 sangsanghay-po-n-ta imagine-try-PRS-DEC
 'I try to imagine whether a new sofa fit into my apartment
 well.'

In English, *think*, *believe*, *hope*, and *fear* were thought to be anti-rogative and that they take only *that* complement clauses and that they don't take complement clauses with polar interrogative complementizer *whether* until White's (2017) corpus data of *think* and *hope* kinds and Peterson's (2017) dissertation example with *imagine whether* (45). The two findings opened a new era of rigorous research in this direction. Naturally, I compare *imagine whether* and the corresponding Korean and Japanese expletive negation complements.

Cross-linguistically, the correspondents of (48) are well attested, as in Italian and Chinese, as in (49').

(49') a. Leo sta immaginando se il divando starebbe bene in salotto.

L is imagining if the sofa would-be well in the living room

'Leo is imagining if the sofa would fit well in the living room.'

(Ivano Ciardelli p.c.)

b. W[˘]o z[˘]ai xi[˘] angxi[˘]ang x[˘] | n sh[˘]af[˘]a sh[˘] | f[˘] ou sh[˘] | h[˘] e w[˘]o

I am imagining new sofa whether fit my

-de k[˘]et[˘] | ng.

I i v i n g - r o o m

'I am imagining whether the new sofa ('d) fit my living room.'

(Linmin Zhang p.c.)

In (49a), the exact Korean translation of (48) in English with the polar interrogative C *whether*, is felt to be slightly more neutral than (46a) with expletive negation in Korean and Japanese. The expletive negation complement with NEG-MOD-Q, composed with the 'imagine' predicate, has positive bias with bouletic modality of 'desire' (and epistemic modality of conjecture).

As we saw with *think*, the attitude predicate *imagine* also occur with *try* and in progressive, as eventive. Both sentences in (50a, b) are positively biased, with (a) optimistic and (b) adversative.

- (50) a. Mary imagined whether her team would win in hope.
b. Mary imagined whether her team would lose in fear.

Now let's turn to the issue of how striking aesthetic effects are created via our imagination. We came to the idea of setting the fiction attitude predicate of IMAGINE with the target complement. Lewis (1978) earlier showed the idea that fictional discourse involves imagining a world where certain propositions hold, where modal operators are used for fictional truth. Metaphors such as (48a) and other figurative expressions or puns like (41b) are fictional languages created.

- (51) a. You are the cream in my coffee. (Grice)
b. Crime Minister Netanyahu
(49) a. I am imagining that you are the cream in my coffee.

Grice views metaphors as a 'parasitic' use of language, derived from conversational maxims (of truth and relevance) via contexts. However, we all appreciate their creative aspects. For Lewis, it is a fictive assertion, invoking a set of worlds where there is a mapping rule f : people $\mapsto$ ingredients in coffee, and the proposition is literally true in those worlds. $\text{FicTrue}_F(p) \Leftrightarrow \forall w \in \text{Best}(F): w \models p$ Where $\text{Best}(F)$ = set of worlds where the fictional rules of discourse hold and which are closest (Lewis similarity metric) to actuality (F = function).

(51a) $\text{FicTrue}_F(\text{Cream}(\text{you}, \text{coffee})) = 1 \Leftrightarrow \forall w \in \text{Best}(F): w \models \text{Cream}(\text{you}, \text{coffee})$. Where F = the imaginative pretense mapping $\text{lover} \rightarrow \text{cream}$, $\text{relationship} \rightarrow \text{coffee}$.

Note the best function employed here and $\Box_F p$ [p true in all fiction-compatible worlds and $\Diamond_F p$ are all weaker than metaphysical necessity @ $\models \Box_F \text{Cream}(\text{you}, \text{coffee})$, which is impossible in the real world, and its existential counterpart.]

‘Crime Minister’ in (53b) is a pun, using the prosodic feature of rhyme, mobilizing imagination so that the expression can be evaluated negatively as a “criminal leader,” juxtaposing with a legitimate counterpart. Its base expression $\text{Prime Minister} = \lambda w. \lambda x. \text{PM}(w)(x)$ changes to ‘Crime Minister’ under imagination: $\text{Crime Minister} = \lambda w. \lambda x. \text{Imagine}(w') [\text{PM}(w')(x) \wedge \text{Criminal}(w')(x)]$, with juxtaposition occurring in the imagined a fictional/conceptual alternative world w' . We can thus model it as an imaginative predicate application.

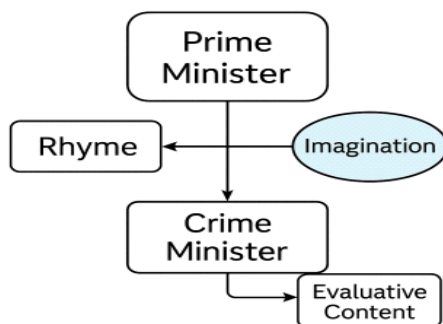


Fig. 7 Creation of Crime Minister

The following personification is another instance of imaginative creation:

(52) Candi-ka apha-hay-yo (잔디가 아파해요).

grass-NOM

pain-feel—DEC-POLITE

‘The lawn is hurting.’ (The grass feels pain)

This sign assigns a human experiential property or ability to feel pain to a non-sentient entity, grass. It violates selectional restrictions but is resolved through an imaginative extension of the discourse context. Formally, $G = \text{Grass}$, $P = \lambda x.\text{feels-pain}(x)$; Literally, $\neg \text{Animate}(G) \Rightarrow P(G)$ undefined or false. Imaginative interpretation, however, introduces a predicate Imagine:

(52') $\text{Imagine}(\lambda w. P(G) \text{ in } w)$ [There is an imagined world w compatible with the speaker's imaginative extension where grass feels pain.]

Fictive Truth (Lewis 1978): $\text{FicTrue}_F(P(G)) \Leftrightarrow \forall w \in \text{Best}(F): w \models P(G)$

Where: F = fictional rules of discourse (grass can feel pain) $\text{Best}(F)$
= worlds closest to actuality except for the fictionality constraint.

This piece of personification sign on the lawn is an indirect illocutionary act of request: ‘Don’t step on the grass and turn around.’

Imaginative Language and Fictive Motion: Joyce (50)

(53) The train was full of fellows: *a long long chocolate train with cream facings*
... *The telegraph poles were passing, passing.* (Joyce 1969: 20, cited
in Brinton 1980, p.374 and Márta Abrusán 2022)

The application of the fictive motion predicate *passing* to inanimate entities such as *telegraph poles* is already metaphorical in character: the poles, in reality static, are construed as undergoing motion. This is a well-attested type of fictive motion (Talmy 2000; Matsumoto 1996) where the observer’s movement is mapped onto otherwise immobile objects. Joyce intensifies this

effect by combining it with metaphorical transformation of the train itself. The descriptive frame shifts from a literal scene—a crowded passenger train—to an imaginative construal in which the train is re-envisioned as an edible confection (*a long long chocolate train with cream facings*).

Mechanisms of Transformation

This transformation exemplifies conceptual blending (Fauconnier & Turner 2002). The literal input space contains the conventional scenario of a passenger train full of people; the second input space is drawn from the domain of desserts (chocolate, cream facings). In the blended space, a new object emerges: an edible train that is simultaneously a vehicle and a cake. The mapping is metaphoric in nature, replacing metal with chocolate, windows with cream, and the physical scene with a gustatory one. The effect is a material metaphor (Lakoff & Johnson 1980), in which an object of transport is reconceived as an object of taste. The rhythmic repetition *long long* and *passing, passing* contributes prosodic iconicity, creating a verbal rhythm that mimics the temporal patterning of train movement.

Formal Representation

(i) Predicate decomposition:

- Literal scene: $\lambda x [\text{train}(x) \wedge \text{full_of}(x, \text{fellows})]$
- Imaginative transformation: $\lambda x [\text{train}(x) \wedge \text{chocolate}(x) \wedge \text{cream_facing}(x)]$
- Dynamic motion: $\exists z [\text{telegraph_pole}(z) \wedge \text{pass}(z)] \wedge \text{iterative}(\text{pass}(z)) \approx \text{pass}(z)^n, \text{ as } n \rightarrow \infty$

(ii) Fictive truth framework (Lewis 1978):

- Define $\text{FicTrue}_F(p)$: $\forall w \in \text{Best}(F) : w \models p$, where F is the set of fictional rules.
- Narrative rule: in Joyce's fictive world, trains may consist of chocolate and cream.
- Result: in all worlds consistent with Joyce's fiction, the proposition

'train = chocolate train with cream facings' holds.

Cognitive and Aesthetic Effects

The effect of such imaginative transformation is one of defamiliarization (Shklovsky 1917): the ordinary experience of train travel is made striking, unfamiliar, and perceptually new. A synesthetic mapping occurs, transferring the sensory qualities of taste and touch (chocolate, cream) onto the visual and auditory experience of the train. This produces heightened sensory engagement and an enriched emotional coloring. The repetition of *passing* in particular foregrounds the fictive construal of telegraph poles as mobile, reinforcing the surreal tone of the description.

Relation to Imagination and Make-Believe

Formally, the transformation can be viewed under the operator 'imagine P' (Lewis 1978; Walton 1990). Readers are invited into a game of make-believe in which the fictional proposition 'the train is chocolate' is accepted as true within the narrative world. The motion of telegraph poles is likewise accepted under fictive rules, even though in reality they do not move. This imaginative stance is distinct from interrogative or disjunctive uses such as 'imagine whether P (or not),' which in inquisitive semantics may undergo expletivization of 'whether or not,' producing hedged positively biased interpretation, composing with a higher predicate of 'imagine.'

In sum, the Joyce passage demonstrates how imaginative creation proceeds:

- (i) A literal perceptual scene (train with passengers) is presented.
- (ii) A metaphoric material transformation occurs (TRAIN → CHOCOLATE DESSERT).
- (iii) Fictive motion applies to external objects (telegraph poles passing).
- (iv) Prosodic repetition evokes rhythm and continuity.
- (v) Formal representation via lambda abstraction and fictive truth captures the shift.

- (vi) Cognitive impact: defamiliarization, synesthetic enrichment, and make-believe participation.

Thus, the passage is an impressive instance of how metaphor, blending, and fictive discourse interact to generate a hybrid cognitive space in which physical reality and edible fantasy co-exist.

6. Concluding Remarks

This study has examined cross-linguistic patterns of factivity alternation, expletive negation with non-canonical *whether*, and imaginative predicates, integrating evidence from English, Korean, Japanese, Chinese, and other languages. Our central claim has been that non-factive alternants of factive verbs can be licensed not only by fixed logical nonveridical operators (negation, modal, conditional, interrogatives, *before*) but also by complement endings (DP vs. Adjunction CP), as in Altaic, and broadly by pragmatic factors — in particular: Distant time expressions (e.g., ‘Six hundred years ago, everyone “knew” the Earth was flat’), which restrict truth to a local historical context while the author’s global common ground maintains the complement’s falsity and its consequent non-factive alternants of various factive kind such as *know* or discovery and emotive factive kind of verbs.

We examined distant temporal contexts as well as false belief contexts (e.g., which allow discovery and emotive factives to appear in non-factive alternants without anomaly. Here, factivity suspension appears to depend on the protagonist’s mental state overly but is ultimately assessed in the author’s global model, rather than through a purely local perspective shift.

By adopting a global view of context, rather than limiting analysis to a sentence-internal perspective shift, we can account for why such pragmatic triggers yield non-factive readings even when local semantics predicts factivity. In Korean, Japanese, and other Altaic languages, this global-local distinction is overtly marked by complement endings: ProFactN for factive alternants versus [+say]reportative for non-factive alternants. In Chinese, the absence of morphological alternation leads to lexical substitution (‘think/believe’ for ‘know/

realize’), further confirming the role of grammar plus focus prosody based on lexical semantic basis via decomposition in encoding the alternation, rather than relying on *believe*-based pragmatic factive view.

A parallel phenomenon appears in expletive negation and non-canonical—*whether* complements. In Korean/Japanese, expletivised negative markers in interrogative complements systematically create a positive bias toward the embedded proposition, functioning as a hedge or politeness device. English *think/believe/hope/fear/imagine+whether* constructions—once marginal—show similar bias-oriented interpretations in corpus and experimental data. We formalised this positive bias in bouletic modal terms, treating expletive negation as a logical trivialisation that removes genuine interrogativity.

Finally, we argued that figurative and imaginative expressions — metaphors, puns, personifications, fictive motion — can be systematically modelled by embedding the target proposition under the predicate *imagine*, interpreted in a Lewisian fictive-truth framework. This approach captures the creativere-mapping of concepts (e.g., ‘You are the cream in my coffee,’ ‘Crime Minister,’ ‘The lawn is hurting’) and the pragmatic effects of defamiliarisation, sensory enrichment, and evaluative stance.

Overall, the phenomena considered here—factivity alternation under logical/semantic or pragmatic licensing, positive-bias interrogative complements via expletivisation, and imaginative predication—converge on a view of attitude semantics that is both structurally constrained and discourse-anchored. The global common ground serves as the ultimate evaluator of truth and presupposition, while local contexts, operators, and pragmatic facts modulate the form and interpretation of embedded propositions. This perspective not only reconciles cross-linguistic variation but also extends naturally to the semantics-pragmatics interface of figurative language.

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초 록

사실성 교체, 허사구문, 그리고 상상세계

이정민

보문유형이 어떻게 상위 태도 술어 유형과 관련되면서 합성성(compositinality)으로 오히려 상위술어를 결정해 나가는지를 보이고자 한다. 인지/의미구조, 통사구조 및 운율구조가 상관관계를 가지면서 언어에 따라 ‘알다’ 등의 인식술어들이 어떻게 사실성 교체(factivity alternation) (FACT-머리어 여부)를 보이는지, 또 교체에서 나온 비사실성 술어와 더불어 ‘생각하다’/‘믿다’ 등의 믿음 관련 (doxastic) 동사들이 어떻게 전달(통보)적(SAY-related) 보문화를(외현적으로라도) 보이는지 고찰한다. 영어는 연산소에 따르나 한국어와 중국어는 이에 무관하여 그 영향권에 들지 않는다. 특히 영어의 표면 현상에 주목한 Perspective Shift (시각 전이) 분석을 비판하고, 말글의 필자(Author)와 공통의 장(Common Ground)을 고려하는 대국적(global) 분석을 주창한다.

또한 ‘상상하다’와 같은 서사와 예술의 허구적 술어의 보문화는 어떻게 되고 영어의 비정규적(non-canonical)(여기서 허사적)whether 보문화는 어떻게 이루어지면서 긍정 편향의 뜻과 디딤말(hedge) 기능을 띄는지 또 이 현상이 어떻게 더 공고한 한국어/일본어의 허사 부정(expletive negation) 현상과 병행적으로 발생-전파되는지 비교 분석한다. 이 특이한 허사적 구문에는 실세계와 구별되는 허구의 보문화를 요하는 ‘상상하다’도 포함된다. 내현적/외현적 머리어나 보문소를 통한 의미 보편소(universal)를 추구하면서 표면상의 중의적 문장이 내포문의 대조적 초점(Rooth 1992)으로 해소되는 과정에 ‘보상적 초점 매기기’도 제안한다.